

Machine Learning Based Prediction of Post-operative Infrarenal Endograft Apposition for Abdominal Aortic Aneurysms

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WHAT THIS PAPER ADDS

This research presents a prediction model based on a pre-operative computed tomography (CT) scan and pararenal aortic shape variation. The model can predict the binarised shortest apposition length (SAL) ($\geq$ or < 10 mm) of an endograft into the infrarenal aortic neck for treatment of an abdominal aortic aneurysm on the first post-operative CT scan, with an accuracy of 77.6%. A SAL < 10 mm has been associated with a higher risk of type Ia endoleak; therefore, this model could help vascular specialists in the pre-operative phase to accurately identify patients who are unlikely to achieve sufficient apposition after endovascular aortic aneurysm repair.

Objective: Challenging infrarenal aortic neck characteristics have been associated with an increased risk of type Ia endoleak after endovascular aneurysm repair (EVAR). Short apposition (< 10 mm circumferential shortest apposition length [SAL]) on the first post-operative computed tomography angiography (CTA) has been associated with type Ia endoleak. Therefore, this study aimed to develop a model to predict post-operative SAL in patients with an abdominal aortic aneurysm based on the pre-operative shape.

Methods: A statistical shape model was developed to obtain principal component scores. The dataset comprised patients treated by standard EVAR without complications ($n = 93$) enriched with patients with a late type Ia endoleak ($n = 54$). The infrarenal SAL was obtained from the first post-operative CTA and subsequently binarised (< 10 mm and ≥ 10 mm). The principal component scores that were statistically different between the SAL groups were used as input for five classification models, and evaluated by means of leave one out cross validation. Area under the receiver operating characteristic curves (AUC), accuracy, sensitivity, and specificity were determined for each classification model.

Results: Of the 147 patients, 24 patients had an infrarenal SAL < 10 mm and 123 patients had a SAL ≥ 10 mm. The gradient boosting model resulted in the highest AUC of 0.77. Using this model, 114 patients (77.6%) were correctly classified; sensitivity (< 10 mm apposition was correctly predicted) and specificity (≥ 10 mm apposition was correctly predicted) were 0.70 and 0.79 based on a threshold of 0.21, respectively.

Conclusion: A model was developed to predict which patients undergoing EVAR will achieve sufficient graft apposition (≥ 10 mm) in the infrarenal aortic neck based on a statistical shape model of pre-operative CTA data. This model can help vascular specialists during the planning phase to accurately identify patients who are unlikely to achieve sufficient apposition after standard EVAR.

Keywords: Abdominal aortic aneurysm, Apposition, Artificial intelligence, Endovascular aneurysm repair, Supervised machine learning

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INTRODUCTION

Challenging infrarenal aortic neck characteristics, such as short length (< 10 mm), severe suprarenal ($> 45^\circ$) and infrarenal

($> 60^\circ$) angulation, conicity, and large diameter (> 30 mm), have been associated with an increased risk of type Ia endoleak (T1aEL) after endovascular aortic aneurysm repair

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(EVAR).^{1–5} Increased physician expertise and improvements in endograft design have led to more patients with high risk anatomical characteristics being treated by EVAR, and a substantial number of these EVAR procedures do not meet the instructions for use (IFU) criteria.^{6,7} However, it can be hard to determine whether the pre-operative aortic geometry of an individual patient is viable for EVAR with a durable outcome, or whether alternative treatments should be performed, such as fenestrated endovascular aortic repair (FEVAR) or open surgery. After treating a patient by EVAR, the real achieved apposition in the aortic neck can be measured, including the shortest apposition length (SAL) on post-operative computed tomography angiography (CTA). SAL is defined as the shortest length of circumferential apposition with the aortic wall, and short apposition length (< 10 mm) on the first post-operative CTA has been associated with T1aEL.⁸

There remains considerable uncertainty about how, and to what extent, specific aortic neck characteristics influence post-EVAR outcomes.^{1,6,9} Previous studies have evaluated the association between a single neck characteristic and post-EVAR complications instead of combining them, which may lead to oversimplification of the complex 3D geometry of the aortic neck.^{3,10–13} To obtain a more comprehensive representation of the aortic shape, the true aortic geometry can be investigated by applying 3D statistical shape modelling (SSM).^{14,15} We recently introduced an SSM of the infrarenal aortic neck of patients with an abdominal aortic aneurysm (AAA). We found that shape descriptors derived from such a model were different for patients treated with EVAR who suffered from a T1aEL compared with controls without complications.¹⁶ The main result of that study was that the pre-operatively created SSM captured information about the identification of a late T1aEL that was complementary to conventional aortic neck measurements. This comparison was followed up in the present study, which aimed to create a pre-operative decision support tool using an SSM of the pararenal aorta that predicts the post-operative shortest endograft apposition length within the aorta based on the pre-operative CTA.

MATERIALS AND METHODS

This retrospective, observational, case control study was conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines and in line with the Declaration of Helsinki.¹⁷ The study was approved by the local Institutional Review Board (registration no. 202100287).

Study design

For a classification model to be able to predict infrarenal apposition, both patients with sufficient and insufficient apposition are needed as input for model training. Aortic segmentations of a large number of patients with and without T1aEL were therefore included and segmented in a standardised manner using a previously validated protocol.¹⁸ The endpoint of the study was to predict whether a patient had a SAL value < 10 mm on the first post-operative

CTA. As patients with a SAL < 10 mm may be at risk of developing a T1aEL, the aim was to include as many patients with a T1aEL as possible.^{8,19} The classification models need such data as input to be able to learn from geometries with insufficient apposition.

Setting

The patient dataset consisted of two groups: (1) patients who were treated by standard EVAR who did not develop a T1aEL during mid to long term CT follow up; and (2) patients who were treated by standard EVAR who developed a late T1aEL, defined as an endoleak that was not yet present on the first post-operative CTA.

As standard of care, patients in this study generally received annual duplex ultrasound follow up. The total imaging follow up, including duplex ultrasound, was used to determine whether a patient developed a late T1aEL. Based on the radiology reports, all patients with a T1aEL confirmed by CT were identified. All post-operative CTAs were checked by two experienced readers (R.C.L.S. and R.Z.) to verify the origin of the endoleak. If classification of the endoleak was in doubt, the scan was discussed within the research group. If after careful consideration there was still doubt about the origin, the patient was excluded; this was the case for one patient, who was then excluded from the study. The first post-operative CTA was used to calculate the SAL in all patients. The follow up protocols are presented in [Supplementary Table S1](#).

Study population

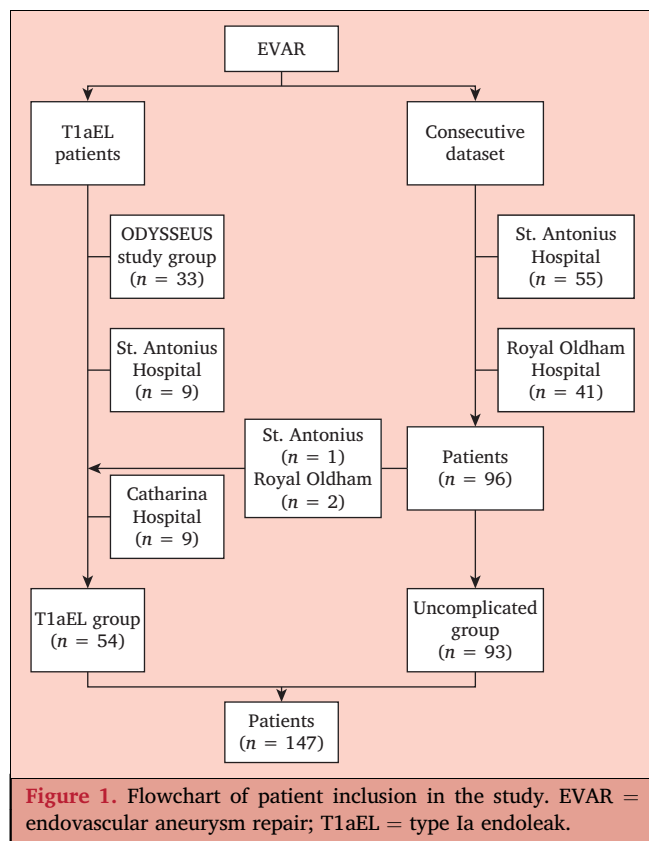
The composition of the dataset evaluated in the present study process is presented in [Figure 1](#).

Patients without a late T1aEL were collected from a dataset of consecutive patients from two hospitals (St. Antonius Hospital, Nieuwegein, the Netherlands; and Royal Oldham Hospital, Manchester, UK).²⁰ Three patients in the dataset were diagnosed with a late T1aEL confirmed by CT and were therefore transferred to the T1aEL group.

EVAR patients who developed a late T1aEL were collected from two different multicentre datasets that included a large number of patients with a late T1aEL confirmed by CT.^{21,22} Inclusion criteria for both groups were: elective repair of AAA treated by standard EVAR; the availability of a pre-operative and 30 day post-operative CTA with sufficient contrast to carry out an aortic segmentation; and no adjunctive proximal fixation, such as additional cuffs or endoanchors. An exclusion criterion was intentional low (> 5 mm) deployment of the endoprosthesis relative to the lowest renal artery. To obtain aortic segmentations, CTAs in the arterial phase and injection of 100 mL of diluted contrast, with a field of view from the coeliac trunk down to the aortic bifurcation, and a slice thickness < 3 mm were needed.

Pre-processing data for classification models

From the pre-operative CTA of each patient, the centre lumen line (CLL), a triangular surface segmentation of the



aortic lumen, and a 3D coordinate of the orifice of the lowest renal artery were semi-automatically obtained by experienced researchers. These researchers followed a training programme, were supervised by a senior researcher with > 10 years of experience with measurements and segmentations, and followed a pre-defined measurement protocol using a 3mensio Vascular Workstation version 10.1 (Pie Medical Imaging B.V., Maastricht, The Netherlands). The CLL, segmentation of the aortic lumen, and 3D coordinate of the orifice of the lowest renal artery were exported to MATLAB 2018a (The MathWorks, Natick, MA, USA) for further analysis.

Post-operative circumferential endograft apposition with the aortic wall was calculated in Vascular Imaging Analysis prototype software (Endovascular Diagnostics B.V., Bussum, The Netherlands) using the CLL, segmentation of the aortic lumen, and 3D coordinates (of both renal arteries, three or four at the proximal end of the endograft fabric and one at

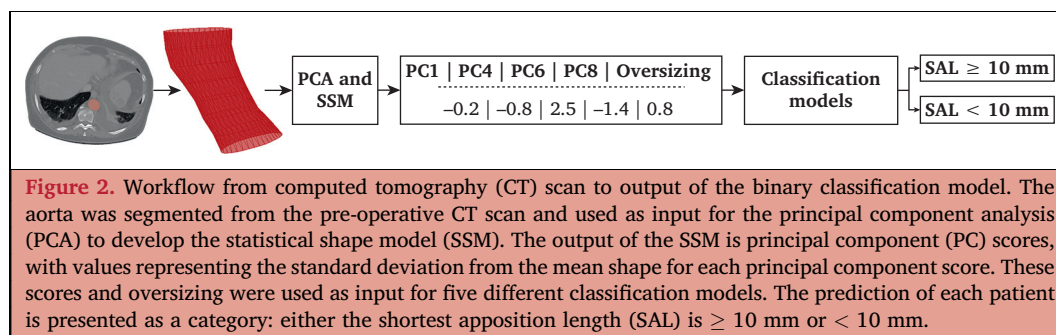
the distal end of circumferential apposition between the endograft and the aorta), according to previously published methods.^{8,18,21} Pre-operative intended oversizing was defined and calculated as the ratio between the pre-operative aortic neck diameter and the nominal endograft diameter.²⁰ The real achieved sealing zone was defined as the length starting at the proximal end of the endograft fabric over which the endograft material is in circumferential apposition to the aortic wall.²³ The SAL was defined as the shortest length between the proximal end of the endograft fabric to the first level where the endograft lost circumferential apposition with the aortic wall.¹⁸

Dedicated software was designed in MATLAB 2018a, with a pipeline for pre-processing of the pre-operative CTA data, registration of the aortic segmentations, principal component analysis (PCA), and creating the SSM. The data were pre-processed for a PCA of the aortic anatomy by means of a parametrisation method, including rigid registration, previously described in more detail.¹⁵ Each segmentation was divided into eight segments, from 5 cm proximal to 2 cm distal to the lowest renal artery coordinate, resulting in eight equidistant contours with 1 cm distance between each contour line. For each contour line, 36 contour points were distributed at equal distance, resulting in a total of 288 contour points per patient. These 288 contour points were used for point to point correspondence of the various patients in this study to conduct PCA and create the SSM. The geometry of each aorta can be described using scores for the principal components. A principal component score is a value that represents the deviation from the mean shape, expressed as a standard deviation. These principal components represent unique anatomical shape variations that are identified by the model.

The workflow is presented in Figure 2, starting with segmentation from a CT scan and resulting in a binary output; the patient is predicted to have either a SAL ≥ 10 mm or < 10 mm on the first post-operative CT scan.

Classification models

Five common classification models were evaluated in this study: (1) logistic regression; (2) random forest; (3) gradient boosting; (4) *k*-nearest neighbours (with number of neighbours *k* equal to nine); and (5) support vector machine.²⁴ These are all supervised learning models, using a labelled dataset to make predictions on new, unseen data. These models use different



mathematical methods to optimise their predictive ability (e.g., using linear or non-linear equations). The scikit-learn (version 1.0.2) package in Python (version 3.9) was used to develop these models. Detailed information about these different classification models can be found in [Supplementary Material S2](#). To compensate for class imbalance between the two SAL groups, a method called synthetic minority oversampling technique (SMOTE) was used.²⁵ By means of this technique, the SAL < 10 mm group was artificially extended to the same number of patients as the SAL ≥ 10 mm group.

Leave one out cross validation was conducted. One patient was taken from the dataset, the SSM was created with this reduced dataset, and the classification models were trained on the reduced dataset. The model was then validated on the patient who was left from the dataset. This process was repeated for all patients in the dataset. Receiver operating characteristic (ROC) curves and area under the ROC curve (AUC) values were computed for all five classification models. A true positive was defined as a patient with < 10 mm predicted apposition and < 10 mm actual apposition; a true negative as ≥ 10 mm predicted apposition and ≥ 10 mm actual apposition; a false positive as < 10 mm predicted apposition but ≥ 10 mm actual apposition; and a false negative as ≥ 10 mm predicted apposition but < 10 mm actual apposition.

Clinically, it is important to correctly predict patients with a SAL < 10 mm (high sensitivity), so a fixed value for sensitivity of 0.70 was chosen. The choice for a threshold with higher sensitivity comes at the cost of lower specificity; however, with the sensitivity fixed at 0.70, the specificity and accuracy were still acceptable (0.78 and 0.79, respectively). Accordingly, the corresponding probability threshold was computed, and this value was used as a cutoff to evaluate whether a patient belonged to the group of patients with a SAL ≥ 10 mm or with a SAL < 10 mm. This threshold was also used to compute the specificity and accuracy of the different models.

Statistical analyses

Data were expressed as the median with 25th and 75th percentiles or as number (%). Differences in the principal component scores of the SSM between the groups with or without SAL ≥ 10 mm were assessed with a Mann–Whitney *U* test. Differences between SAL values and pre-operative intended oversizing between the uncomplicated and T1aEL groups were assessed with a Mann–Whitney *U* test. DeLong's test was used to evaluate whether the performance of the model with the highest AUC was significantly different from the other models.^{26,27} IBM SPSS Statistics, version 28.0 (IBM Corp, Armonk, NY, USA) and Python (version 3.10) were used for statistical analysis. A *p* value of < .050 was considered statistically significant.

RESULTS

Study population

Of the 147 patients, 126 (85.7%) were men, and the mean age was 74 ± 8 years. Twenty six patients (17.7%) had a

pre-operative neck length < 10 mm. The median slice thickness of the CTAs was 1.0 (1.0, 1.5) mm. Twenty four patients had an infrarenal SAL of < 10 mm with a median measured value of 4.6 (0.9, 7.7) mm, and 123 patients had a SAL of ≥ 10 mm with a median measured value of 22.6 (16.1, 31.5) mm. The median follow up was 47 (15.8, 64) months for the SAL < 10 mm and 41 (19.3, 62) months for the SAL ≥ 10 mm group. Fourteen patients (58%) in the SAL < 10 mm group and 114 (92.7%) in the SAL ≥ 10 mm group were treated within IFU, respectively.

The measured median SAL of the uncomplicated EVAR group was 22.6 (15.7, 31.7) mm vs. 14.6 (5.9, 23.0) mm in the T1aEL group (*p* < .001). Four patients (4%) had a SAL < 10 mm in the uncomplicated group, whereas 20 patients (37%) had a SAL < 10 mm in the T1aEL group. The median pre-operative intended oversizing in the uncomplicated EVAR group was 17% (11.0, 25.0%) vs. 13.3% (7.9, 28.0%) in the T1aEL group (*p* = .46).

Pre-processing of the data for classification models

The Mann–Whitney *U* test showed that principal components 1, 4, 6, and 8 were statistically significantly different between the groups with or without SAL measuring ≥ 10 mm ([Fig. 3](#)). These four principal components and oversizing were used as input for the classification models.

Classification models

The ROC curves of the five classification models are presented in [Figure 4](#). The highest AUC value of 0.77 was obtained by the gradient boosting classification model. With a probability cutoff threshold of 0.21, 114 patients (77.6%) were correctly classified. The sensitivity (correct classification of < 10 mm SAL) was 0.70 and specificity (correct classification of ≥ 10 mm SAL) was 0.79. The AUC, accuracy, sensitivity, and specificity of the four other classification models, and the results of the DeLong's test, are presented in [Table 1](#). The true positive, true negative, false positive, and false negative rates of the gradient boosting model with the corresponding measured SAL value are presented in [Figure 5](#). Median measured SAL values for each of the four subgroups are displayed, as well as the distribution of T1aEL and uncomplicated patients among the four subgroups.

Patient examples

Four patient examples are provided in [Figure 6](#) to show the predictive ability of the gradient boosting model in a single patient. Representative patient examples were presented for the true positive, true negative, false positive, and false negative groups.

DISCUSSION

Five classification models based on machine learning were developed for patients who underwent elective EVAR. The best performing model, the gradient boosting model, had an accuracy of 77.6% for predicting sufficient SAL of the

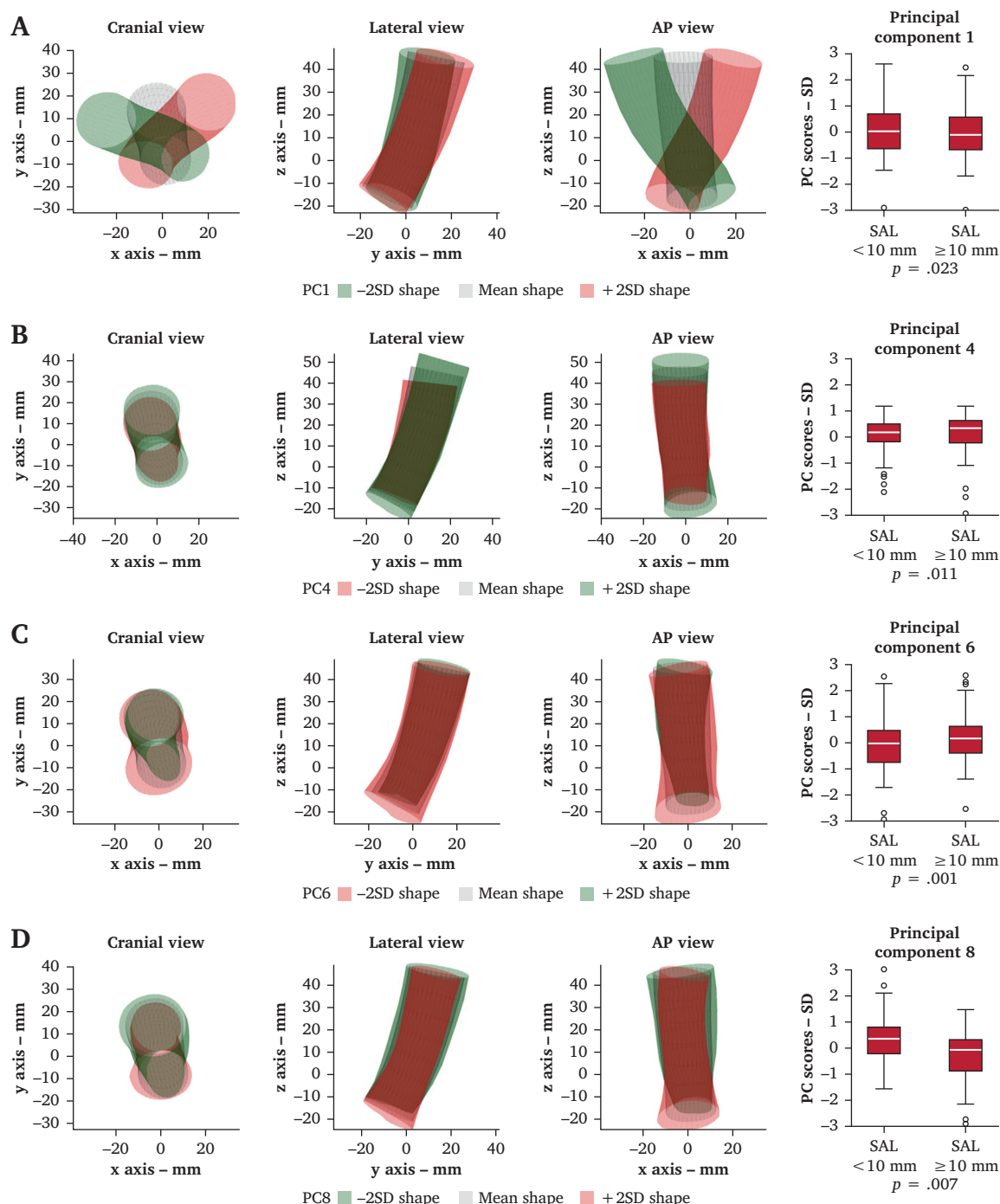
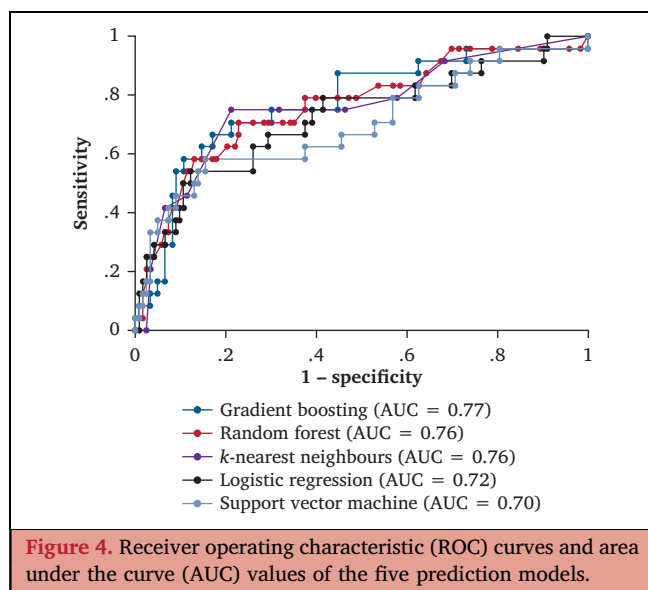


Figure 3. Statistical shape model with the four principal components (PCs) that were statistically significantly different between the groups with or without shortest apposition length (SAL) of < 10 mm. Green shapes are associated with SAL ≥ 10 mm, whereas red shapes are associated with SAL < 10 mm. (A) PC1 describes 54% of the total shape variation and includes left–right deflection of the aorta. (B) PC4 describes 5% of the total shape variation and includes conicity combined with angulation of the aorta. (C) PC6 describes 2% of the total shape variation and includes conicity with angulation of the aorta. (D) PC8 describes 1% of the total shape variation and includes a taper at the level of the lowest renal artery. AP = anteroposterior view; SD = standard deviation.

endograft in the infrarenal aorta. Consequently, such a classification model can be used as a quantitative measure to assess risk of apposition length < 10 mm, based on pre-operative anatomical information. The model shows the direct influence of the combination of challenging neck characteristics on the anticipated achieved apposition. The

model can be used during treatment planning as an addition to conventional measurements of the anatomy and the IFU to determine suitability for EVAR.

This study has shown the ability of the SSM to capture pre-operative anatomical shape variation that is predictive of post-operative SAL. With the probability threshold set to



0.21, the gradient boosting model reached high specificity and accuracy values. As the sensitivity value was chosen, the specificity and accuracy values were lower compared with a higher threshold. However, a false positive prediction may prompt the surgeon to perform a FEVAR procedure in a neck that would have been suitable for standard EVAR. Because of the relatively high specificity, we expect that this would not regularly occur with the current models. On the other hand, as a result of a false negative prediction, standard EVAR might be chosen as a treatment option while this results in an actual apposition of < 10 mm, leaving the patient at risk of a T1aEL. Furthermore, we did not investigate whether SAL that was falsely predicted to be < 10 mm eventually decreased to < 10 mm, and SAL that was falsely predicted to be ≥ 10 mm eventually increased to sufficient length. Even though principal components 2, 3, and 5 mainly contributed to the total shape variation in the dataset, these principal components are not necessarily clinically relevant in determining the prediction of the shortest circumferential apposition. As a result, these principal components were not statistically significantly different between the two groups and were not included as input for the classification models.

A segment of the aorta between 5 cm proximal to 2 cm distal to the lowest renal artery was used for this SSM. Other segmentations of the aorta would result in a different SSM and may therefore give other results. For the distal cutoff, we chose to cover the area that is most relevant for the proximal seal without including too much of the aneurysm (2 cm distal to the lowest renal artery). For the proximal cutoff, we chose to include a part of the supra-renal aorta that is generally relevant for FEVAR (up to 5 cm proximal to the lowest renal artery). This was done to create a generic model for all patients with AAA that can be appended with FEVAR patients in the future research.

The two groups in this study were defined by a cutoff SAL of 10 mm; this value is defined as an inadequate seal in the European Society for Vascular Surgery (ESVS) 2024 clinical practice guidelines.²⁸ Previous studies have also shown that patients with a SAL < 10 mm are at high risk of AAA related adverse events after EVAR and development of a T1aEL.^{8,19}

The study by Liang *et al.*²⁹ is another example of research that combined the output of an SSM as input for prediction models. They investigated the relationship between shape features and predicted rupture risk with support vector machine classifiers.²⁹ Wang *et al.*³⁰ developed different prediction models based on anatomical features, and predicted several EVAR related outcomes, including type I and III endoleaks, re-intervention, and rupture. Their support vector machine morphological feature model had the best predictive performance, with an AUC of 0.76.³⁰ Moreover, Avril *et al.* described the use of computational modelling of stent graft deployment and optimisation of EVAR procedures.³¹ An efficient classification tool was implemented in the present study to obtain predictions of apposition in a computationally faster manner compared with the models described above.

Limitations

The first limitation of this study is related to the study data, as they came from different datasets. Although the EVAR group comprised consecutive patients, the dataset was enriched with non-consecutive patients with a late T1aEL from different centres. This was the only way to collect enough patients with a T1aEL while not having to segment hundreds of aortas, but it may have created selection bias.

Table 1. Results of the five classification models.

Classification model	AUC	Accuracy	Sensitivity	Specificity	<i>p</i> value
Gradient boosting	0.77	0.78	0.70	0.79	BPM
Random forest	0.76	0.64	0.70	0.77	.69
<i>k</i> -nearest neighbours	0.76	0.48	0.70	0.79	.78
Logistic regression	0.72	0.30	0.70	0.63	.40
Support vector machine	0.70	0.55	0.70	0.47	.13

A fixed value for sensitivity was chosen (0.70), from which the corresponding threshold was defined. With this threshold (0.21), the accuracy and specificity for each model was computed. *p* values indicate whether the performance of the model with the highest AUC (gradient boosting) was statistically different from the other models, evaluated by the DeLong's test. A *p* value of $< .050$ was considered statistically significant. AUC = area under the receiver operating characteristic curve; BPM = best performing model.

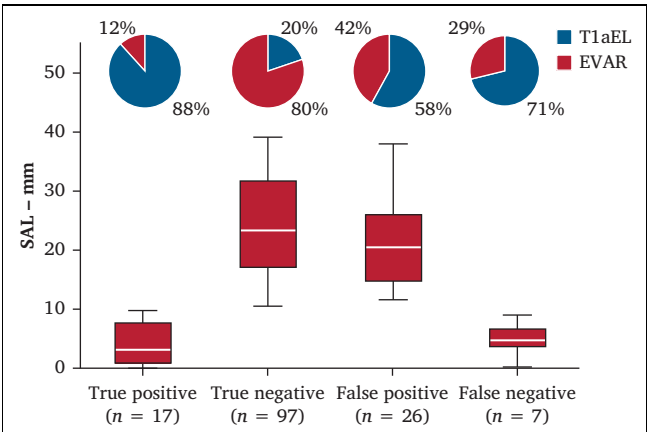


Figure 5. Boxplots of the measured shortest apposition length (SAL) on the first post-operative computed tomography scan, displayed for each of the four subgroups true positive, true negative, false positive, and false negative, based on the gradient boosting model. A combination of principal components 1, 4, 6, and 8, and oversizing was used as input in the model. The row of pie charts represents the distribution of the type Ia endoleak (T1aEL) and endovascular aneurysm repair (EVAR) groups among the four subgroups.

We believe that this did not bias the outcome of the model because the SAL was determined on the first post-operative CTA scan, which was made within six weeks after the operation as standard of care in all participating centres. Moreover, only patients with a late T1aEL were appended, owing to the inclusion criteria from the previous studies. Therefore, patients with a T1aEL on the angiogram or first post-operative CT scan were not included in this model. Because of the retrospective nature of this study, it was unknown whether the choice for a particular endograft was influenced by the anatomy. The first identification of the T1aEL was based on the radiology reports; therefore, there may have been a risk of misclassification or missing data. However, the experienced readers carried out core lab analyses on all CTAs; therefore, minimal risk of misclassification or missing a T1aEL was assumed.

Acquisition of aortic segmentation and placement of a coordinate of the lowest renal artery in a vascular workstation, next to creation of a CLL, might not be part of routine pre-operative EVAR planning in all medical centres. Obtaining these parameters as input for the classification model may take extra time, and expertise is required to accurately determine all parameters. However, in the last year automatic algorithms have been developed (e.g., automatic extraction of a CLL or automatic segmentations), which might be helpful when implemented in clinical practice.^{32,33}

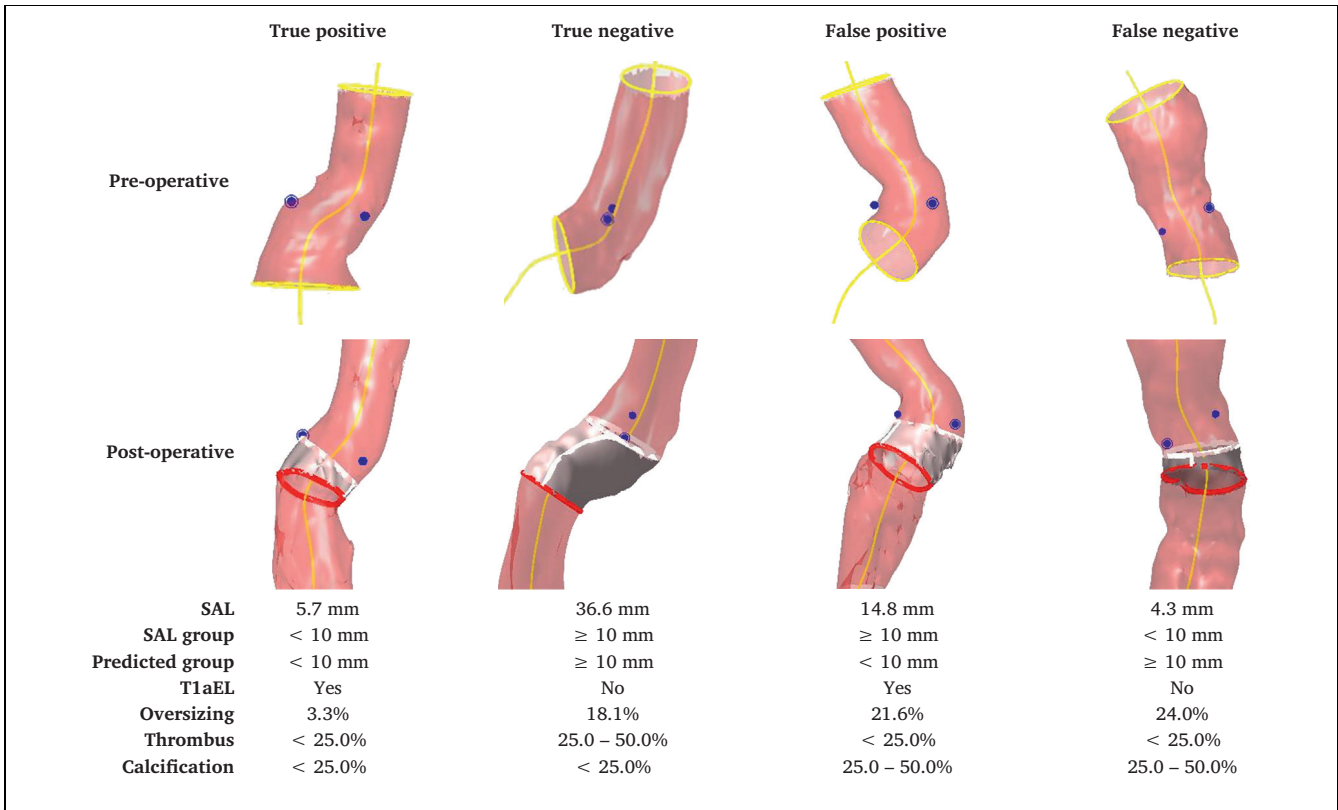


Figure 6. Patient examples showing the pre-operative aortic shape and the actual shortest apposition length (SAL) on the first post-operative computed tomography angiogram. Pre- and post-operative segmentations of the aorta are shown with two blue coordinates, representing the renal artery orifices. The yellow contour lines (pre-operative) represent the boundaries of the segmented aorta that was input for the statistical shape model. The grey surface area (post-operative) represents the real achieved sealing zone. The white line covering the grey surface represents the SAL. T1aEL = type Ia endoleak.

Multiple researchers semi-automatically acquired the aortic segmentations, which may have resulted in segmentation bias. However, potential differences in segmentation would have little influence on the total shape variation, and, therefore, present in higher principal components. As the classification models in this study included four principal components that describe larger shape variations, it is assumed that the influence of interobserver variability is low.

Future perspectives

This study evaluated the SAL on the first post-operative CT scan. Change in SAL during follow up is equally important for the development of a late T1aEL. Therefore, in future research the models may also be trained to predict change in the SAL during follow up. The influence of thrombus and calcification was not evaluated in the classification models in this study, as lumen segmentations were used to develop the SSM. Thrombus might play a role in achieving sufficient apposition, but this could not be evaluated in the present study because thrombus information was only available as a binominal parameter. Use of outer wall segmentations, or addition of thrombus segmentation in the model, might further improve the accuracy of the classification models.^{33–35} Not only geometric parameters may be predictive of apposition, but also patient demographics, comorbidity, aortic wall elasticity, and stent graft deployment might also influence the achieved SAL. As a future perspective, this model should be externally validated with a consecutive patient series that includes patients with an early T1aEL before it can be used as a patient specific virtual stenting tool in clinical practice.

Conclusion

A model to predict which EVAR patients will achieve sufficient graft apposition (≥ 10 mm) in the infrarenal aortic neck based on a statistical shape model of pre-operative CTA data was developed. This model can help vascular specialists during the planning phase to accurately identify patients who are unlikely to achieve sufficient apposition after standard EVAR.

CONFLICT OF INTEREST

Jean-Paul de Vries and Richte Schuurmann are co-founders of the company Endovascular Diagnostics B.V., which holds patent rights over the software used to determine endograft apposition and position.

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APPENDIX A. SUPPLEMENTARY DATA

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ejvs.2024.07.003>.

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